FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject	:	PROBABILITY & STATISTICAL DATA ANALYSIS (SECI 1143)
Task	:	Chapter 7 (Correlation and Regression) - **85 marks**
Due Date	:	**1 week after release (please refer to the section's lecturer)**

INSTRUCTION:

1. This is a **GROUP** assignment. Please clearly write your member's **NAME & MATRIC NUMBER** on the front page of the submission.
2. This assignment contributes to **5%** of overall course marks.
3. Only **HANDWRITTEN** submission is accepted:
   a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
   b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
   c. For answers that need to draw graphs, using graph paper is optional. You can use plain paper.
   d. Round your answers to **THREE** decimal places.
4. Submission **online** or in **hard copy (please refer to the section's lecturer).**

QUESTION 1 **(15 MARKS)**

A biscuit production manager wants to investigate the linear relationship between the number of items produced and the production cost. To pursue his/her objective, the manager recorded the data on the number of items produced per day and the production cost per day (in thousands of Malaysian Ringgit).

Table 1 shows the data collected for 12 consecutive days.

Days	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Number of items, (x)	26	44	53	29	77	80	20	40	67	86	17	61
Production cost, (y)	42	60	69	47	91	98	39	55	85	104	37	77

a. Name the **TWO (2)** most popular correlation coefficient. (2 marks)
b. Calculate the correlation coefficient, *r*. (11 marks)

1a) the two most popular corelation coefficiants are Pearson's product-moment corelation coefficiant and Spearman's rho rank corelation coefficiant.

b)

number of items, x	production cost, y	xy	x^2	y^2
26	42	1092	676	1764
44	60	2640	1936	3600
53	69	3657	2809	4761
29	47	1363	841	2209
77	91	7007	5929	8281
80	98	7840	6400	9604
20	39	780	400	1521
40	55	2200	1600	3025
67	85	5695	4489	7225
86	104	8944	7396	10816
17	37	629	289	1369
61	77	4697	3721	5929
$\Sigma = 600$	$\Sigma = 804$	$\Sigma = 46544$	$\Sigma = 36486$	$\Sigma = 60104$

$n = 12$

$$r = \frac{\Sigma xy - (\Sigma x \Sigma y)/n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2/n][(\Sigma y^2) - (\Sigma y)^2/n]}}$$

$$= \frac{46544 - 600(804)/12}{\sqrt{[36486 - 600^2/12][60104 - 804^2/12]}}$$

$$= \frac{6344}{\sqrt{40446696}}$$

$$r = 0.998$$

c) $r = 0.998$, it is extremely near to the $r = 1$.
It is a relatively strong positive linear association between the number of items produced (x) and the production cost (y)

2a)

weight of plastic usage (x)	size of household (y)	xy	x^2	y^2
0.27	2	0.54	0.0729	4
1.41	3	4.23	1.9881	9
2.19	3	6.57	4.7961	9
2.83	6	16.98	8.0089	36
2.19	4	8.76	4.7961	16
1.81	2	3.62	3.2761	4
0.85	1	0.85	0.7225	1
3.05	5	15.25	9.3025	25
$\Sigma = 14.6$	$\Sigma = 26$	$\Sigma = 56.8$	$\Sigma = 32.9632$	$\Sigma = 104$

$n = 8.$

$$r = \frac{\Sigma xy - (\Sigma x \Sigma y)/n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2/n][(\Sigma y^2) - (\Sigma y)^2/n]}}$$

$$= \frac{56.8 - 14.6(26)/8}{\sqrt{[32.9632) - (14.6)^2/8][104 - (26^2)/8]}}$$

$$= \frac{9.36}{\sqrt{123.2049}}$$

$$r = 0.843$$

$\therefore$ $r = 0.843$, it is close to $r = 1$.

It is relatively a strong positive linear association between the weight of plastic usage (x) and the size of household (y)

b) $\alpha = 0.05$

$H_0 : \rho = 0$

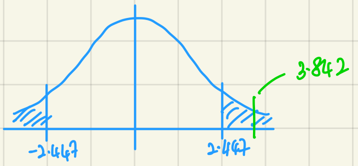
$H_1 : \rho \neq 0$

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

$$= \frac{0.8432}{\sqrt{\frac{1-0.8432^2}{8-2}}}$$

$$t = 3.842$$

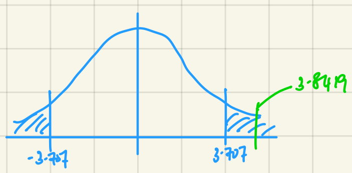
$$t_{0.025,6} = 2.447$$



Since $3.842 > 2.447$ $(t > t_{0.025,6})$. We reject the null hypothesis.
There is a sufficient evidence of a linear relationship between the weight of plastic usage (x) and the size of household (y) at the 95% confidence level.

c) $\alpha = 0.01$

$$t_{0.005,6} = 3.707$$



Since $3.842 > 3.707$ $(t > t_{0.005,6})$. We reject the null hypothesis.
There is a sufficient evidence of a linear relationship between the weight of plastic usage (x) and the size of household (y) at the 99% confidence level.

c. Based on the correlation coefficient, r obtained, make a conclusion on the linear relationship between the number of items produced and the production cost

(2 marks)

$$r = \frac{\sum xy - \left(\sum x \sum y\right)/n}{\sqrt{[(\sum x^2) - (\sum x)^2/n][(\sum y^2) - (\sum y)^2/n]}}$$

QUESTION 2 **(10 MARKS)**

A local authority is going to study the relationship between the size of households and the daily plastic usage consumed by them. The results of a sampling given below in Table 5 indicate some values of *X* (weight of plastic usage) and *Y* (the size of household).

Table 2: Weight of plastic usage (*X*) and size of household (*Y*)

X	0.27	1.41	2.19	2.83	2.19	1.81	0.85	3.05
Y	2	3	3	6	4	2	1	5

a. From the above sampling, find the value of correlation coefficient *r*. (5 marks)

b. Using the same sample data set above, conduct the hypothesis testing to know whether the variables *X* and *Y* correlate using a 95% confidence level. (3 marks)

c. Find out if the decision in (ii) may change if you increase the confidence level to 99%.

(2 marks)

QUESTION 3 **(25 MARKS)**

A social media analytics company is investigating the relationships between various factors to understand how they influence the engagement on posts. The company has collected data on the following variables (Table 3) for 6 different social media posts:

- **Number of Likes**: The number of likes received by the post.
- **Number of Comments**: The number of comments received by the post.
- **Number of Shares**: The number of times the post was shared.
- **Post Length**: The length of the post in characters.
- **Engagement Score**: A score (from 1 to 100) representing the overall engagement on the post.
- **Sentiment Score**: An ordinal variable representing the sentiment of the post (1: Very Negative, 2: Negative, 3: Neutral, 4: Positive, 5: Very Positive).

Table 3: Factors influencing engagement on posts

Post ID	Likes	Comments	Shares	Post Length	Engagement Score	Sentiment Score
1	150	20	30	200	85	4
2	100	10	20	100	70	3
3	200	25	40	250	90	5
4	80	8	15	150	60	2
5	170	22	35	220	88	5
6	120	15	25	180	75	3

a) Name two variables whose relationship can be expressed by Spearman's Rho Rank correlation coefficient. (2 marks)

b) Name two variables whose relationship can be expressed by the Pearson correlation coefficient. (2 marks)

c) Compute the Spearman correlation coefficients for the same variables in (a). (8 marks)

d) Identify and compare the strongest positive and negative correlations from:
   a. Likes and comments
   b. Likes and share
   c. Engagement Score and Sentiment Score (6 marks)

e) Test the null hypothesis that there is no correlation between 'Engagement Score' and 'Post Length' against the alternative hypothesis that there is a significant correlation. Use a significance level of 0.05 and provide the test statistics and p-values. (7 marks)

QUESTION 4 **(15 MARKS)**

Last year, five randomly selected students took a math aptitude test before they began their statistics course. The Statistics Department would like to analyze the relationship of the data, make predictions using the regression equation and validate the regression equation. The data of the math aptitude test score and corresponding statistics grade are as shown in Table 4 below:

Table 4: Score on Math and Statistics grade

Score on math aptitude test (x)	Statistics grade (y)
95	85
85	95
80	70
70	75
65	70

Answers at 2 decimal places.

a. Calculate $\sum x$, $\sum y$, $\sum xy$ and $\sum x^2$ (4 marks)

b. Calculate value of b1 and b0 (2 marks)

c. Based on answers in (a) and (b), what linear regression equation best predicts statistics performance, based on math aptitude scores? (2 marks)

d. Which graph represents the regression equation in (c)? (1 mark)

e. If a student made a 60 on the aptitude test, what grade would we expect her to make in statistics? (2 marks)

f. Find the value of SSR, SST and R^2. (3 marks)

g. Based on your answer in (f), how well does the regression equation fit the data? (1 mark)

QUESTION 5 **(20 MARKS)**

A software development company wants to predict the time required to complete new software projects based on the estimated lines of code (LOC). The company has collected data on the development time (in weeks) and LOC for previous projects. Using this data presented in Table 5, answer the following questions in 2 decimal places:

Table 5: Length of code and development time

Project ID	LOC (K)	Development Time (weeks)
1	150	40
2	100	35
3	200	50
4	80	30
5	170	45
6	120	38
7	160	42
8	90	28
9	250	55
10	130	37

a. Plot a scatter plot of the data with LOC on the x-axis and Development Time on the y-axis. (3 marks)

b. Calculate the correlation coefficient between LOC and Development Time. (7 marks)

c. Fit a simple linear regression model using LOC as the independent variable and Development Time as the dependent variable. Provide the regression equation and interpret the coefficients. (5 marks)

d. Calculate the R-squared value and interpret it. (3 marks)

e. Use the regression model to predict the development time for a new project with an estimated 180K LOC. (2 marks)